

Name : Bisma fatima

Class : CS-A 24

Roll No : F24605036

Course : Artificial Intelligence

Code : CS-280

Lecturer : Ma'am Noushin Saba

Assignment 03

Question No 01

Your task is to schedule CS classes that meet Monday, Wed and Friday. There are 5 classes that meet on these days and 3 professors who will teach these classes. You are constrained by the fact that each professor can only teach one class at a time.

a) Formulate this as a CSP. Identify:

i. Variables (one per class)

- X_1 : Professor assigned to class 1 - Intro to programming
(8:00 - 9:00 am)
- X_2 : Professor assigned to class 2 - Intro to AI
(8:30 - 9:30)
- X_3 : Professor assigned to class 3 - Information Retrieval
(9:00 - 10:00 am)
- X_4 : Professor assigned to class 4 - Data Science
(9:30 - 10:30 am)
- X_5 : Professor assigned to class 5 - Computer Networks
(10:00 - 11:00 am)

ii. Domains (professors available for each class)

- $X_1 \in \{C\}$
- $X_2 \in \{B, C\}$
- $X_3 \in \{A, B, C\}$

$$\bullet X_4 \in \{A, B, C\}$$

$$\bullet X_5 \in \{B, C\}$$

iii. Constraints

- Availability constraint: Professors can only teach classes they are listed for.
- Non-overlap constraint: A professor cannot teach two classes that overlap in time.

b) Draw Constraint graph showing overlap.

Classes overlap if their time intervals intersect:

- Class 1 overlaps class 2
- class 2 overlaps with class 1 and class 3
- Class 3 overlaps with class 2 and class 4
- Class 4 overlaps with class 3 and class 5
- class 5 overlaps with class 4.

So the graph edges are:

$$\bullet (1-2), (2-3), (3-4), (4-5)$$

This form a chain-like constraint graph.

c) Domains after applying availability

Same as a(ii):

$$X_1 = \{C\}$$

$$\bullet X_2 = \{B, C\}$$

$$\bullet X_3 = \{A, B, C\}$$

$$\bullet X_4 = \{A, B, C\}$$

$$\bullet X_5 = \{B, C\}$$

d) One Valid Solution

We need to assign professors so that no one teaches overlapping classes.

- Assign $X_1 = C$ (only option)
- Assign $X_2 = B$ as "C overlap with Class 1 (X_1)"
- X_3 overlaps with class 2 (B). choose $X_3 = A$
- X_4 overlap with class 3 (A), choose $X_4 = C$
- X_5 overlap with class 4 (C), X_5 must be B.

Final Assignments:

- class 1 $\rightarrow$ professor C
- class 2 $\rightarrow$ professor B
- class 3 $\rightarrow$ professor A
- class 4 $\rightarrow$ professor C
- class 5 $\rightarrow$ professor B

All constraint satisfied -